How does land use regulation influence spatial residential segregation?

Southern Methodist University, Department of Economics*

October 9, 2020

Abstract

This paper examines how residential land use regulation affects residential segregation by income across different spatial scales in metropolitan areas in the United States. Residential segregation by income may reduce access to opportunity for lower income households. Different spatial scales account for the population composition of neighborhoods within different distances of a given neighborhood. To address the endogeneity of land use regulation, 2-stage least squares regressions use plausibly exogenous instrumental variables that proxy metropolitan area regulatory tendency: protective inspection expenditure share and number of streams. The paper finds that increasing local land use regulation intensity, measured by the Wharton Residential Land Use Regulation Index (WRLURI), reduces income segregation across spatial scales. Nevertheless, results for WRLURI subindices provide weak but suggestive evidence that state involvement and approval delay drive negative results while other types of regulation, like residential density restrictions (minimum lot size restrictions), are associated with higher income segregation at lower spatial levels.

JEL classification: R

Keywords: Land Use Regulation, Zoning, Spatial, Economic Geography, Urban Economics

1 Introduction

Residential segregation by income has been rising since the 1970s in the United States with a brief pause in the 1990s and wide variation among metropolitan areas (Reardon, Bischoff, Owens, and Townsend 2018). The percent of families living in neighborhoods with median income outside 50% to 150% of the metro area median rose from 15% in 1970 to 34% in 2012 (Reardon and Bischoff 2016, Galster and Sharkey 2017). Residential segregation is higher and has risen most among households with children, impeding intergenerational social and economic mobility (Owens 2016, Reardon et al. 2018).

Land use regulation may contribute to residential segregation. Residential land use regulations can protect a community's public amenities but slow housing stock growth and raise housing costs, contributing

*Email: bstutts@smu.edu. Acknowledgements: James Lake, Daniel Millimet and Klaus Desmet

to skill sorting between and possibly within metropolitan areas (Glaesar and Gyourko 2014, Ganong and Shoag 2017). Restricting the quantity of housing within a high-demand area limits residential access to those most willing to pay. Higher income households outbid lower income households for limited housing within highly-regulated high-opportunity metropolitan areas. This may concentrate high-income households in high-demand neighborhoods and sort lower-income households into lower opportunity or more peripheral neighborhoods within those metro areas as well as into separate metro areas entirely.

Local zoning laws across the United States restrict residential density to varying degrees. While municipalities typically set land use rules affecting income segregation at the municipality scale and smaller geographic units, their policies may also affect housing markets and the distribution of people in other jurisdictions across the broader economic region, approximated by a Metropolitan Statistical Area (MSA). Without density restrictions, rising land values per housing unit in high-demand neighborhoods encourage higher density development until the premium per housing unit disappears. Density limits preserve housing unit premiums in high-demand neighborhoods, geographically separating the housing market by income level.

This paper investigates how land use regulation influences residential segregation by income within metropolitan areas, accounting for spatial distance between neighborhoods.[1]I hypothesize that metropolitan density restrictions will increase spatial segregation and high-income residential concentration by making high-demand areas highly supply inelastic.

Examining segregation between neighborhoods within metropolitan areas requires a definition of neighborhood boundaries. A neighborhood alludes to a local spatial area of regular physical social interaction, but no uniform definition of boundaries exists. Segregation measures commonly use census tracts or block groups to proxy neighborhood boundaries. Chetty, Hendren, Kline, and Saez (2014) observes sharp contrasts in economic mobility even at the block level in some metropolitan areas, suggesting segregation may matter at many spatial scales.

Traditional segregation measures compare within-neighborhood characteristics between neighborhoods within the metropolitan area without accounting for distance between neighborhoods. Effective segregation may be less severe when high and low income neighborhoods are close together than when they are distant. Ranking metropolitan areas by residential segregation changes depending on the choice of measure and

[1]I use Metropolitan Areas from 1999 defined by the Office of Management and Budget (OMB) under a complex set of standards (see: https://www.govinfo.gov/content/pkg/FR-1998-12-21/pdf/98-33676.pdf#). Typically, a metropolitan area includes central counties (cities/towns only in New England) containing most of a city or urbanized area of at least 50,000 people and outlying counties (cities/towns only in New England) with any of the following: (1) at least 50% of employed residents commuting into the central counties and at least 25 people per sqmi; (2) 40%+ commuting and 35+ per sqmi or 5000+ residents live in qualifier urbanized area; (3) 25%+ commuting and 50+ per sqmi or two of following: (a) 35+ per sqmi, (b) 35% urban population, (c) 5000+ in qualifier urbanized area; (4) 15%+ commuting (or workers commuting from central counties is 15%+ and all workers commuting to and from central counties if 20%+ of employed residents) and two of following: (a) 60+ per sqmi, (b) 35% urban population, (c) 20%+ population growth between last decennial censuses, (d) 5000+ in qualifer urbanized area; (5) 2500+ residents live in central city.

level of analysis: block, block group, census tract, etc. (Lee, Reardon, Firebaugh, Farrell, Matthews, and O'Sullivan 2008). This paper's primary segregration measure, the spatial information theory index from Reardon and O'Sullivan (2004), accounts for varying degrees of spatial proximity between neighborhoods.

Rothwell and Massey (2010) and Lens and Monkonnen (2016) offer empirical evidence suggesting density-restricting land use regulation increases segregation. However, they use suspect identification strategies and don't account for the socioeconomic composition of nearby neighborhoods. This paper follows closely the specification in Rothwell and Massey (2010) and Lens and Monkonnen (2016). Like Lens and Monkonnen (2016), I use the Wharton Residential Land Use Regulation Index (WRLURI), instead of the Pendall land use regulation survey that Rothwell and Massey (2010) uses, to increase the number of observations. I extend their work with more convincing instrumental variables (IVs) and additional measures of segregation that capture the geographic distance between neighborhoods, using Census block group as the unit of analysis for constructing the segregation measures.

Primary specifications measure MSA residential segregation by income using the spatial rank-order information theory index, from Reardon, Firebaugh, O'Sullivan, and Matthews (2006), using 2012-2016 5-year ACS block-level Census data and the NBER block group distance database for the 2010 US Census. I regress this measure on MSA-level averages of municipal WRLURI scores and separately on each component index, including sociodemographic controls. Because historical segregation patterns and other factors likely influence both 2005 land use regulation and 2012-2016 segregation, I instrument for land use regulation using the 1982 share of protective inspection and regulation expenditures from US Census of Governments and 1991 stream count from the United States Geological Survey (USGS) (Dawkins 2005, Saiz 2011). I also regress the distance-decay interaction index and spatial information theory index of black-white segregation on WRLURI and its components.

Primary 2-stage least squares (2SLS) specification results indicate land use regulation, measured by WRLURI score, reduces MSA segregation by income under a range of proximity-weighted measures that incorporate geographic distance between block groups. Varying coefficient signs by WRLURI component suggest differential effects by the type of land use regulation. The Local Zoning Approval Index has a positive and significant association with segregation at all but the smallest spatial scales. The Density Restrictiveness Index is positively associated with segregation but only significant in ordinary least squares (OLS). The main contribution of this paper is demonstrating how the effects of land use regulation vary significantly with income segregation measures that account for spatial proximity between neighborhoods. As spatial scale increases, the variance in income segregation compresses and coefficient magnitudes for WRLURI attenuate almost proportionally but peaking at the 4000m spatial scale.

2 Literature Review

2.1 Importance of Segregation

The United States exhibits wide geographic disparities both between and within metro areas in upward social and economic mobility (Intrator, Tannen, and Massey 2016; Galster and Sharkey 2017; Lens 2017). Neighborhoods play a crucial role in education and economic outcomes (Chetty et al. 2014; Chetty and Hendren 2018). Rothwell and Massey (2014) estimate the lifetime earnings gap between a top- and bottom-quartile census tract is $910,000. Housing costs reduce economic mobility by making it harder for low-income households to live in high-opportunity neighborhoods and still finance education and save (Chetty, Friedman, Hendren, Jones, and Porter 2020).

Residential income segregation deprives these families not only of higher quality local amenities but also of diverse social networks that can share specialized expertise and career help. Hence, segregation reduces upward economic mobility and increases social and economic inequality (Rothwell and Massey 2014; Ewing, Hamidi, Grace, and Wei 2016; Reardon and Bischoff 2011; Owens, Reardon, and Jencks 2016; Fogli and Guerreri 2019; Jargowsky 2018; Chetty et al. 2014; Chetty and Hendren 2018).

These unequal opportunities deprive poorer children regardless of their potential. This perpetuates social ills in poor neighborhoods and lowers overall productivity and innovation. Hence, economic segregation worsens outcomes in income, education, health, and public safety, slowing long run economic growth (Jargowsky and Wheeler 2017; Acs, Pendall, Treskon, and Khare 2017; Dreier, Mollenkopf, and Swanstrom 2014).

2.2 Measuring Segregation

2.2.1 Dimensions of Segregation and Measure Criteria

Massey and Denton (1988) identify five dimensions of segregation: evenness, exposure, concentration, centralization, and clustering. Reardon and O'Sullivan (2004) condense all five dimensions into two broader dimensions: spatial evenness and spatial exposure. Spatial evenness reflects how similarly different population groups distribute across residential space within a metro area. Spatial exposure reflects how prevalent members of one population group are in the residential space near a member of another group.

James and Taeuber (1985), Reardon and Firebaugh (2002), and Reardon (2011) describe 12 desirable features for segregation indices: organizational equivalence, size invariance, composition invariance, transfers, exchanges, ordered exchanges, scale interpretability, rank-preserving scale invariance, additive organizational decomposability, additive group decomposability, spatial adaptability, and threshold invariance (Royuela and Vargas 2010). Organizational equivalence implies splitting or merging sub-areas

with the same group shares does not affect segregation. Size invariance implies mulitplying each group's population in each sub-area by the same constant does not affect segregation. Composition invariance implies multiplying the population of the same group in each sub-area by the same constant does not affect segregation. Transfers implies moving an individual from a sub-area with a higher share of the individual's group to a sub-area with a lower share reduces segregation. Exchanges implies sub-areas trading individuals of different groups such that sub-area shares become more similar reduces segregation. Ordered exchanges implies sub-areas trading individuals of more different groups changes segregation more than does trading individuals of less different groups.

Scale interpretability implies segregation is minimized if and only if group shares in all sub-areas equal group shares overall and segregation is maximized if and only one group occupies each sub-area. Rank-perserving scale invariance implies independent of income inequality and comparable across place and time regardless of units of income or cost of living differences. Additive organizational decomposability implies segregation can partition into between- and within-area components. Additive group decomposability implies segregation can partition into between- and within-group components. Spatial adaptability implies segregation accounts for spatial patterns of individuals. Threshold invariance implies insensitivity to choice of thresholds dilineating ordered groups.

The dissimilarity index, a popular evenness measure, compares a group's population share in each sub-area unit to that in the MSA overall. The Dissimilarity Index of Poverty (D) is:

$$D = \sum_j \frac{P_j \times |\frac{LP_j}{P_j} - \frac{LP}{P}|}{2 \times LP \times (1 - \frac{LP_j}{P_j})} \tag{1}$$

where j is sub-area unit j, P is population, and LP is population below poverty. When the number of groups exceeds two, the dissimilarity index fails composition invariance and transfers. It only weakly satisfies exchanges (and transfers in two-group case) and fails additive organizational decomposability and additive group decomposability.

Rothwell and Massey (2010) use an another evenness measure, a population-weighted Neighborhood Gini Index of inequality in median household income across sub-areas:

$$Gini = \frac{N+1}{N-1} - \frac{2 \times [\sum_j (O_j \times P_j \times M_j)]}{N \times (N-1) \times [\sum_j (P_j \times M_j)]} \tag{2}$$

where N is number of sub-areas, M_j is median income in subarea j, and O_j is rank of sub-area j by median income. *Gini*'s sensitivity to N fails organizational equivalence. *Gini* fails transfers (if population-weighted), compositional invariance, and additive decomposability.

My main specifications use Reardon et al. (2006)'s spatial rank-order information theory index, a

weighted average of pairwise segregation values, based on the entropy concept (Reardon 2011). It satisfies rank-preserving scale invariance, scale interpretability, spatial adaptability, organizational equivalence, size invariance, exchanges, ordered exchanges, and additive organizational decomposability. It doesn't satisfy composition invariance, ordered group decomposability, or threshold invariance. But Reardon (2011) demonstrates its robustness to income category thresholds for US MSAs if using all Census income bins and to the set of income bins if bins span most of the income distribution.

The spatial information theory index captures spatial evenness. It builds on the aspatial Theil's H Information Theory Index from Reardon (2011) and Reardon et al. (2018). For aspatial indexes with binary categories like affluent population HP and poor population LP and sub-area unit j:

$$H = \sum_j [\frac{HP_j + LP_j}{HP + LP} \times \frac{(v - v_j)}{v}] \tag{3}$$

where

$$v = -[\frac{HP}{HP+LP} \times log_2 \frac{HP}{HP+LP} + (1 - \frac{HP}{HP+LP}) \times log_2(1 - \frac{HP}{HP+LP})] \text{ if } 0 < \frac{HP}{HP+LP} < 1, 0 \text{ else.}$$

It sums these values of each block group j in the MSA weighted by the block group's share of the MSA's affluent and poor population.

Lens and Monkkonen (2016) use the Rank-Order Information Theory Index:

$$H_R = 2ln(2) \int_0^1 -[\frac{P_m}{P} log_2 \frac{P_m}{P} + (1 - \frac{P_m}{P}) log_2(1 - \frac{P_m}{P})] \times (1 - \frac{\sum_{j=1}^{J} P_j \times [\frac{P_{jm}}{P_j} log_2 \frac{P_{jm}}{P_j} + (1 - \frac{P_{jm}}{P_j}) log_2(1 - \frac{P_{jm}}{P_j})]}{P \times [\frac{P_m}{P} log_2 \frac{P_m}{P} + (1 - \frac{P_m}{P}) log_2(1 - \frac{P_m}{P})]} dm \tag{4}$$

a weighted average of pairwise segregation indices using population counts of household income bins (Reardon 2011). The index calculates how closely each block group's population shares above and below each income bin match those of the overall MSA. It sums these values of each block group in the MSA weighted by the block group's share of the MSA's population.

Spatial Rank-Order Information Theory Indexes account for the population composition of nearby sub-area units up to some radius distance r (e.g. 500m) (Reardon et al. 2006). Spatial-adjusted rank-order bins use a biweight proximity function:

$$w_{jk} = [1 - (\frac{d_{jk}}{r})^2]^2 \text{ if } d_{jk} < r, 0 \text{ else}$$

for every combination of sub-area units j and k. For each group m in each sub-area j, there is a spatially-weighted local population composition (LPC) function:

$$LPC_{jm} = \frac{\int_{k \in J} P_{km} \times w_{jk} dk}{\int_{k \in J} P_k \times w_{jk} dk}$$

that divides the sum of all these weighted nearby-sub-area populations. The sum for all sub-areas j in MSA J of $P_{km} \times w_{jk}$ takes the place of the aspatial population count bins by MSA with order zero to calculate MSA-level spatial rank-order information theory indexes. Spatial information theory indexes likewise compute spatial information theory measures of binary-group segregation using the spatial-weighted LPC function's $P_{km} * w_{jk}$ in place of the aspatial binary-group population counts in the information theory equation. The Spatial Multigroup Information Theory Index denoted by $\tilde{H}$ is:

$$\tilde{H} = 1 - \frac{\int_{j \in J} P_j \times [-\sum_{m=1}^{M} LPC_{jm} log_M(LPC_{jm})] dj}{P \times [-\sum_{m=1}^{M} \frac{P_m}{P} log_M(\frac{P_m}{P})]} \tag{5}$$

where M is the number of population groups m. Spatial Rank-Order Information Theory Index denoted by $\tilde{H}_R$ is:

$$\tilde{H}_R = 2ln(2) \int_0^1 -[\frac{P_m}{P} log_2 \frac{P_m}{P} + (1 - \frac{P_m}{P}) log_2 (1 - \frac{P_m}{P})] \times [1 - \frac{\int_{j \in J} P_j \times [LPC_{jm} log_2(LPC_{jm}) + (1 - LPC_{jm}) log_2(1 - LPC_{jm})] dj}{P \times [\frac{P_m}{P} log_2 \frac{P_m}{P} + (1 - \frac{P_m}{P}) log_2 (1 - \frac{P_m}{P})]}] dm \tag{6}$$

for income threshold percentile m between 0 and 1.

Exposure reflects the degree to which different groups reside in common areas. For example, the aspatial interaction index measures the likelihood a randomly-selected person in an MSA belongs to a different population group as another person randomly-selected from their sub-area. The Interaction Index (of Poor with Affluent) denoted by I is:

$$I = \sum_j (\frac{LP_j}{LP} \times \frac{HP_j}{P_j}) \tag{7}$$

for sub-area j, population P, population in poverty LP, and affluent population HP.

Interaction indices depend on relative populations of social groups measured, making it hard to compare segregation across MSAs with different income compositions (Royuela and Vargas 2010). Rothwell and Massey (2010)'s exposure index controls for an MSA's overall level of inequality to compare across MSAs. The Exposure (of Poor to Affluent) Index denoted by X is:

$$X = \frac{\sum_j (\frac{LP_j}{P_j} \times \frac{HP_j}{P_j})}{\sum_j \frac{LP_j}{P_j}} \tag{8}$$

for sub-area j, population P, population in poverty LP, and affluent population HP.

Distance decay isolation and interaction indexes capture spatial exposure but also inequality, muddying cross-MSA comparisons. The Distance Decay Interaction Index of Poor with Affluent denoted by DDI is:

$$DDI = \sum_j [\frac{LP_j}{LP} \times \sum_k (\frac{P_k^{-d_{jk}}}{\sum_j P_j^{-d_{jk}}} \times \frac{HP_k}{P_k})] \tag{9}$$

where d_{jk} is distance in miles between the internal points of sub-area units j and k.

2.2.2 Geographic Unit of Analysis

Spatial scales matter for segregation. Aspatial measures don't capture proximity between sub-areas, resulting in the checkerboard problem (Lens 2017, Reardon and O'Sullivan 2004). A metro area with segregated neighborhoods dispersed evenly throughout the metro area, like red and black squares on a checkerboard, would score no differently than a metro area with equally segregated neighborhoods concentrated, like a board all red on the left and all black on the right. Aspatial measures also suffer from the modifiable areal unit problem: sensitivity to the choice of sub-area unit land area size and boundaries to the extent proportions of population groups in nearby sub-areas vary.

Tract-based aspatial measures make less dense areas appear less segregated than under an egocentric spatial approach with individualized person-specific neighborhoods, attenuating the observed effect of density restrictions (Lee et al. 2019).[2] Tract-based measures score similarly to an egocentric approach to the spatial information theory index of residential segregation at 500-2000m scales but differ at 4000m scale (Lee et al. 2008). Tract and block group land area shrinks with population density, which may reflect less dense suburban neighborhoods covering greater geographic distance due to greater driving prevalence and speed but miss greater regular interaction among a larger population in denser neighborhoods. Spatial weights capture the proximity of nearby areas beyond sub-area unit boundaries (Reardon, Firebaugh, Matthews, and O'Sullivan 2008; Reardon, Matthews, O'Sullivan, Lee, Firebaugh, Farrell, and Bischoff 2008; Lee et al. 2008; Reardon and Bischoff 2011; Barrett, Farrell, Reardon, and Matthews 2019).

The most relevant geographic scale for residential income segregation varies by individual and environmental charactersitics. Smaller spatial scales may matter more for those with restricted mobility like young children and the elderly. Chetty et al. (2020) find characteristics of blocks only within a mile of a child's block can help explain a child's future earnings. Intermediate spatial scales like middle and

[2] See: www.pop.psu.edu/mss

high school zones may be more important for teenagers. Larger distances may matter more for mobile working-age adults.

Direct measures more precisely capture segregation in school, crime, and proximity to jobs. School districts affect the school a child attends. Municipal boundaries affect police, fire, and other municipal services received. Crime, social networks, and job accessibility have less discrete geographic scales. Residential segregation contributes to these major disparities as well as peer group and social network segregation that are harder to directly measure with public data. Spatial-weighted measures of residential segregation offer a less precise but more holistic starting point to assess socioeconomic segregation overall.

The spatial information theory index is sensitive to modifying the radius and spatial-weighting but is non-increasing in scale (Reardon, Matthews, et al. 2008). A 500-meter radius approximates the area of daily pedestrian activity. A 4000-meter radius approximates a high school zone. Relative segregation between MSAs varies with scale, but correlation across scales usually exceeds .90. Adopting the spatial information theory index from Reardon and O'Sullivan (2004) allows this paper to account for the proximity of people in one block group to people in nearby block groups and explore how land use regulation might affect spatial evenness at different geographic scales.

2.3 Land Use Regulation and Segregation

Empirical findings show land use restrictions, particularly on density, increase residential segregation by inome and race (Berry 2001, Rothwell and Massey (2009), Massey and Rugh (2014), Rothwell and Massey 2010, Lens and Monkkonen 2016). The Wharton Index closely correlates with a Census tract's ratio of housing costs to upward mobility (Chetty et al. 2020) However, results vary by the measure of segregation and land use regulation. My contribution, closely building on Lens and Monkonnen (2016) and Rothwell and Massey (2010), applies distance-based spatial segregation measures that better reflect effective segregation of economic opportunity and an alternative identification approach to address endogeneity.

Rothwell and Massey (2010) find density zoning restrictions increase residential income segregation in US MSAs but find no evidence of an effect for other land use regulations or broader indices like WRLURI. They measure segregation via a Neighborhood Gini coefficient (equation 2), an exposure index of poor to affluent (equation 8), and a poverty dissimilarity index (equation 1). Rothwell and Massey measure maximum permited dwelling units per acre and other zoning rules using Pendall, Puentes, and Martin (2006)'s 2003 survey of local land use regulations in 1,677 out of roughly 5,000 total jurisdictions in the 50 largest MSAs in the United States. Rothwell and Massey's 2SLS estimates use year of statehood and MSA population density in 1910 as IVs and control for socioeconomic and environmental characteristics.

Lens and Monkkonen (2016) find lot size minimums increase segregation of the high and middle income

but not low income. Municipal review process complexity and local political pressure to restrict land use also increase segregation. State involvement in land use regulation reduces income segregation. They measure 2010 MSA segregation using the aspatial Reardon and O'Sullivan (2004) rank-order information theory index (equation 4) that compares the percent difference in MSA income diversity and a population-weighted sum of each block group's income diversity. They run OLS with controls on MSAs with over 500,000 people using the 2005 Wharton Residential Land Use Regulation Index (WRLURI) survey of 11 categories of local land use regulations from Gyourko, Saiz, and Summers (2008) for 1,904 jurisdictions in 293 US MSAs.

3 Methods and Data

3.1 Conceptual Framework

Land use rules, especially those restricting population density, may affect residential segregation by income. Local residential zoning restrictions can effectively set minimum housing costs that limit strictly zoned areas to highest incomes. Metro-wide residential zoning restrictions raise fixed costs of housing, further circumscribing lower-income housing options. However, if residential preferences favor sorting by income in equilibrium, then, especially if neighhorhood quality changes over time, raising permitting costs may reduce residential re-sorting by income. The size and direction of any effect of land use rules depends on the nature of residential preferences.

The Schelling (1971) model represents residential segregation driven by willingness-to-live amongst peers where full residential segregation is the unique stable equilibrium under even a slight preference for living amongst peers. Building off Schelling's checkerboard model of segregation, imagine a checkerboard with rich and poor people buying a location on checkerboard without any restrictions on number of people per squares and no preferences over squares or neighbors. Rich and poor will randomly disperse across the board in relatively equal proportions. Without locational or peer preferences, restricting density would not increase segregation, except to increase the average distance between everyone on the checkerboard. Introducing an exogenous public good in specific squares would induce everyone to move to these higher quality squares. In this case, restricting density forces some people to live outside the high quality square, resulting in a premium to buy access to the high quality square. Income segregation emerges with the rich concentrated in the high quality squares if the rich have higher willingness to pay for the public good.

Adding congestion costs would, without density restrictions, result in some people choosing to live outside the high quality neighborhoods when the disutility of congestion in the high quality neighborhood exceeds utility gained from the high quality neighborhood. The rich (poor) sort out of the high quality

square if the ratio of willingness to pay for neighborhood quality to congestion costs is lower among the rich (poor). Equal congestion costs alone would not produce segregation by a specific group. With congestion externalities, homeowners in the high quality square have an incentive to impose density restrictions both to raise their home-equity and to limit the disutility they experience from congestion. Again, density restrictions provide a separating equilibrium of rich (poor) in high quality square and poor (rich) outside if willingness to pay for neighborhood quality net of congestion costs is higher among rich (poor), except now poor (rich) will spread evenly across low quality squares to minimize density. Congestion utility might more realistically have a quadratic shape, so high quality square owners would set density restriction to maximize utility by maximizing price per square meter, and poor would spread evenly across low quality squares at utility-maximizing densities.

Permitting restrictions that raise development time and cost in high quality squares sufficiently high (regardless of cost in low quality squares) would result in only the rich moving from lower- to higher-quality squares due to an inelastic housing stock. However, if square quality changes over time, permitting costs may slow neighborhood sorting by income and potentially lower segregation at any point in time.

If a high quality square improves the quality of nearby squares by the degree of proximity, then people will prefer to live not just in the high quality square but will prefer to live closer rather than further from the high quality square. Square owners near a high quality square will have an incentive to collectively restrict near-high quality square density to capture rent. A universal density restriction will create quality premiums diminishing in distance from high quality square. Rich most willing to pay for quality will concentrate in squares closest to a high quality square, segregating from poor who will encircle rich in next closest set of squares to set of squares housing all the rich. A partial density restriction produces the same result except poor will concentrate in squares at least as close as closest unrestricted square until congestion costs exceed the square's benefits.

Sufficiently high permitting costs will result in all rich (most willing to pay) moving into high quality and near high quality squares with the poor dispersed across board and unable to afford to move closer (any starting in highest quality squares will prefer to sell square to rich who can afford to develop property or will leave as rent becomes unaffordable). However, if square quality changes over time, permitting costs may slow this process of sorting by income, preserving mixed-income squares.

3.2 Empirical Framework

The empirical framework closely follows Rothwell and Massey (2010) and Lens and Monkonnen (2016). The empirical specification regresses MSA-level segregation variables from 2016 5-year ACS on the 2005 Wharton Residential Land Use Regulation Index (WRLURI). Because WRLURI uses 2000 MSA bound-

aries, my controls are mostly from the 2000 Census to limit errors cross-walking from 2016 to 2000 MSA boundaries to the segregation and Gini inequality variables. I first replicate Lens and Monkonnen (2016) main regressions except with ACS 2012-2016 segregation data instead of 2010 data. I then add instrumental variables and spatial scales to the Lens and Monkonnen (2016) regressions.

I use the model:

$$S_i = \beta_0 + \beta_1 R_i + X_i^{'} \beta_2 + u_i$$

where i is MSA (2000 Census MSA/PMSA). S is a measure of segregation. Types of measures include spatial and aspatial versions of the interaction and information theory indices. For each of the spatial information theory indexes, I run specifications with measures at 500m, 1000m, 2000m, and 4000m radii. My preferred measure is the spatial information theory index.

R is the score for the WRLURI or one of its components. X is a vector of controls: the natural log of the MSA's total population (2000 Census), the Gini Index of Income Inequality (ACS 2016 5-year), the share of households with annual household income less than $20,000 (2000 Census), the share of households with annual household income at least $60,000 (2000 Census), the share of the total population who is Black alone or Hispanic alone (2000 Census), and the number of general purpose municipality governments (e.g. cities and townships) in 2002.

3.2.1 Identification

Land use regulation may be an endogenous regressor for residential segregation because past segregation and other variables may influence both an MSA's land use policy and current segregation. Measuring land use regulation with a 2005 index of municipal land use regulation requires some means of identification to address endogeneity concerns. This paper seeks to address this concern through the use of instrumental variables that plausibly affect land use regulation but not segregation.

Rothwell and Massey (2010) instrument for land use regulation with year of statehood and 1910 population density, which the authors suggest proxy for MSAs with more established suburban settlements that would typically restrict land use more than would new settlements. Rothwell and Massey (2010) observe that metropolitan areas with later statehood year tend to have had less time for rural settlements to form around cities. They hypothesize older rural settlements are more likely to have established political coalitions opposing new development, citing literature supporting this claim. Such settlements may be more prevelant in historically denser metropolitan areas. They observe zoning restrictiveness measured by a zoning survey has a strong negative correlation with statehood year and positive correlation with 1910 population density, consistent with this hypothesis. To be valid, 1910 population density and state age

must only affect segregation via channels captured by variables included in the regression model.

While statehood and 1910 population density mostly predate the adoption of zoning and other modern land use regulations, these instruments may not be valid if MSAs in older states or with higher 1910 population densities are systematically more or less prone to segregation through channels outside land use policy such as their effect on modern population density.[3] Statehood year may be invalid if it proxies for historical racial segregation. 1910 population density of MSAs may be subject to bias from endogenous MSA boundaries. For instance, an MSA in California includes predominantly rural Kern County, one of the geographically largest counties in the US. By contrast, New England MSAs define boundaries by municipalities not counties so are less likely to include rural areas and appear denser than in other states.

Instead, my preferred MSA-level instrumental variables for WRLURI are 1982 protective inspection and regulation expenditure share and Dawkins (2005)'s number of stream segments in the 1991 USGS hydrography layer from the 1:2,000,000 DLG data. Dawkins (2005) and Hoxby (2000) employed number of streams as an instrument for municipal competition on segregation. They note that streams often formed municipal boundaries, especially in the 19th century when even smaller streams posed significant transportation barriers. Thus, areas with a greater number of streams would be more likely to have a greater number of municipal jurisdictions. This greater number of jurisdictions would also create greater incentive for local regulation attempting to capture local land rents within an MSA and free ride as much as possible on surrounding neighborhoods by restricting their own land use. Cutler and Glaeser (1997) find no significant correlation between number of streams and segregation after controlling for the correlation between number of jurisdictions and segregation.

Additionally, I use Saiz (2010)'s local public protective inspection and regulation expenditures as a share of local government revenues from the Census 1982 State and Local Government Finance Data. Saiz (2010) argues this choice of local government spending allocation may proxy for local preference for regulation independent of preferences for segregation and uses it to instrument for WRLURI in housing elasticity regressions. While protective expenditure share correlates positively with WRLURI, its validity as an instrument rests on its spending share not relating to segregation via channels not captured in the regression model. We thus assume the historical spatial distribution of high and low income within the metropolitan area doesn't influence the preference or willingness to pay for protective spending, even though segregated and mixed-income jurisdictions may have very different fiscal capacities.

[3] Rothwell and Massey (2010) include population density as a control in their initial regression with extensive controls before pairing down to their preferred parsimonious regression.

3.3 Data

3.3.1 Dependent Variables: Aspatial and Spatial Segregation Measures

This paper uses ACS 5-year 2016 block-group and NBER Census Block Group Distance Database data crosswalked to 1999 MSA boundaries of 295 metropolitan areas to create MSA-level segregation measures used for the dependent variables.[4] Each regression employs either a version of the information theory index to capture evenness or interaction index to capture exposure, or their spatial counterparts to capture spatial evenness and spatial exposure, respectively.

Interaction Index of Poor with Affluent **(I):**

I is equation 7 where j is block group j, P is population, LP is population below the poverty threshold, and HP is population with income at least 200% of poverty threshold.

Interaction Index of Low with High Income **($IHHI$):**

$IHHI$ is equation 7 where j is block group j, P is number of households, LP is number of households with income less than $25,000, and HP is number of households with income at least $100,000.

Rank-Order Information Theory Index (Theil's H) for Poverty**:** I use the STATA "rankseg" command to create a rank-order information theory index (equation 4) using population counts of bins for the following percentage ranges of the poverty threshold: 0-49%, 50-99%, 100-124%, 125-149%, 150-184%, 185-199%, and 200%+. I set order at zero.

Rank-Order Information Theory Index for Income**:** I use the STATA "rankseg" command to create a rank-order information theory index (equation 4) using number of household counts of bins for the following household income ranges: $0-9,999, $10,000-14,999, $15,000-19,999, $20,000-24,999, $25,000-29,999, $30,000-34,999, $35,000-39,999, $40,000-44,999, $45,000-49,999, $50,000-59,999, $60,000-74,999, $75,000-99,999, $100,000-124,999, $125,000-149,999, $150,000-199,999, and $200,000+. I set order at zero.

Black-White Information Theory Index for Race**:** For the simpler non-rank information theory indexes, I create my own code for a Theil's H (equation 5), attempting to closely follow Reardon and

[4] ACS 5-year 2016 block group population counts are not adjusted to ensure they match official population estimates. US Census encourages using ACS block group data only to compare percentages and averages, not population or sub-group population counts. However, segregation measures rely on these population counts. A future revision will use 2020 Census data for block group population counts, but for now the segregation measures are susceptible to significant measurement error. See: https://www.census.gov/programs-surveys/acs/guidance/handbooks/researchers.html

Townsend who supply STATA with a "seg" command. Alternatively, I could use the "seg" command but current results reflect calculations by my own STATA construction of the following equation 3 from Reardon (2011) and Reardon et al. (2018) for racial categories White non-Hispanic in place of HP and Black non-Hispanic LP.

Distance Decay Interaction Index of Poor with Affluent **(DDI):** DDI is equation 9 where P is population, LP is population below the poverty threshold, HP is population with income at least 200% of poverty threshold, and d_{jk} is distance in miles between the internal points of block groups j and k.

Distance Decay Interaction Index of Low with High Income Households **($DDIHHI$):** $DDIHHI$ is equation 9 where P is number of households, LP is number of households with income less than $25,000, HP is number of households with income at least $100,000, and d_{jk} is distance in miles between the internal points of block groups j and k.

Spatial Rank-Order Information Theory Indexes for Poverty and for Income**:** Following Reardon et al (2006), I create separate indexes (equation 6) that account for the population composition of nearby block groups up to each of the following radii r: 500m, 1000m, 2000m, and 4000m. The "rankseg" command doesn't include a spatial option, so I create spatial-adjusted rank-order bins. First, I create the biweight proximity function: $w_{jk} = (1 - (d_{jk}/r)^2)^2 \; if \; d_{jk} < r, \; 0 \; else$, for every combination of block groups j and k. For each group m in each block group j, I create the spatially-weighted local population composition (LPC) function: $LPC_{jm} = \int P_{km} * w_{jk} dk / \int P_k * w_{jk} dk$, that is I divide the sum of all these weighted nearby-block-group populations. I use the sum for all block groups j of $P_{km} * w_{jk}$ in place of the aspatial population count bins and run "rankseg" command by MSA with order zero to create MSA-level spatial rank-order information theory indexes. This method overweights block groups surrounded by denser populations, so I will re-run these estimates using my own code for future revisions.

Spatial Black-White Information Theory Indexes for Race**:** Similarly to spatial rank-order information theory indexes above, I follow same procedure to calculate spatial versions of the Black-White information theory measures of racial segregation above using the spatial-weighted LPC function's $P_{km} * w_{jk}$ in place of the aspatial racial population counts in the information theory equation.

3.3.2 Key Explanatory Variables: Land Use Regulation Measures

The 2005 Wharton Residential Land Use Regulation Index (WRLURI) proxies for land use regulation. Gyourko, Saiz, and Summers (2008) discusses their 2005 survey of municipal Planning Directors or Chief

Administrative Officers on residential land use regulation in 2,649 municipalities in the United States. Gyourko provides weights to aggregate the index into a 1999-MSA-level dataset, which I use (dropping municipalities outside year-1999 MSAs).[5] WRLURI decomposes into the following 11 subindices.

Local Political Pressure Index (LPPI) assesses the importance for "residential building activity and growth management" of local and county government, community pressure, city budget constraints, school crowding, opposition to growth, and local land conservation initiatives. Local political pressures may be a response to recent growth and diversification of the local population that I expect would slow future growth and integration. County government importance may affect segregation at a larger spatial scale than city council, city budget constraints, city council opposition to growth, increasing segregation primarily between municipalities. School crowding may increase segregation between school districts and school zones. I expect community pressure and citizen opposition to growth to primarily increase segregation at the smallest neighborhood spatial scales. Local land conservation ballot measures may reduce intra-jurisdiction segregation if open space protection makes affluent more willing to live in denser mixed-income neighborhoods, but otherwise may increase intra- or inter-jurisdictional segregation as lower income families who can't afford housing adjacent to open space amenities sort into outlying communities by proximity to amenities.

State Political Involvement Index (SPII) reflects both the adoption of new statewide land use restrictions 1995-2005 and state legislative involvement in local "residential building activity and growth management." New state land use restrictions likely slow not only housing growth but also neighborhood sorting, implying ambiguous segregation effects, because state governments may favor more inclusionary restrictions and statewide restrictions raise development costs in both new and old neighborhoods. While perhaps more likely to ease segregation at larger spatial scales, I expect the spatial scale of any segregation effects to depend on the specific state building codes, environmental rules, and growth management policies the state implements.

State Court Involvement Index (SCII) captures the tendency of state appellate courts to uphold municipal land use regulations. I expect this empowers municipalities to impose exclusionary zoning, reinforcing segregation across all spatial levels.

Local Zoning Approval Index (LZAI) reflects the number of local bodies required to approve any zoning change request. Local Planning Approval Index (LPAI) reflects the number of local bodies required to approve a project. Local Assembly Index (LAI) is an indicator for certain New England localities that require popular approval at an open town meeting for any zoning change. Added barriers to new and higher density housing will likely reinforce segregation but could potentially slow neighborhood transitions toward

[5] WRLURI data are available as of April 2020 at http://real-faculty.wharton.upenn.edu/gyourko/land-use-survey/

segregation, particularly at smaller spatial scales.

Supply Restriction Index (SRI) reflects the presence of statutory limits on the number of building permits, number of units authorized per year, total number of multifamily dwellings, and number of units per multifamily building. I expect restricting total multifamily dwellings and units per multifamily building, by limiting housing density and mix, will increase macro-segregation in growing metros between high and low demand regions of the MSA. Limiting building permits and units authorized per year likely slows both new housing growth and neighborhood sorting, implying an ambiguous expected effect on segregation.

Density Restriction Index (DRI) is an indicator for having a one-acre minimum lot size requirement. I expect minimum lot size will increase segregation by limiting both maximum housing density and affordable housing options.

Open Space Index (OSI) is an indicator for having open space requirements, requiring lots to dedicate some portion for open space limits development options. But neighborhoods that preserve open space amenities without explicitly limiting housing may be more amenable places for the affluent to live and still permit lower income housing options, albeit raising cost of providing low-income options on lots with open space rules.

Exactions Index (EI) is an indicator for having exactions on developers for a share of infrastructure costs related to the development. Exactions reduce new housing growth, but paying the cost of additional infrastructure may deter new developments segregated from other neighborhoods, with effects likely at larger spatial scales.

Approval Delay Index (ADI) captures the expected number of months between application and receipt of building permit. A slower permitting process raises fixed costs making some affordable housing projects financially infeasible, likely increasing segregation, but it could also slow neighborhood sorting, potentially extending a neighborhood transition period between segregated equilibrium states.

3.3.3 Control Variables

MSA-Level US Census 2000 Control Variables: From the decennial US Census 2000 SF1 and SF3 data files at the PMSA/MSA level, I obtain the following metropolitan-level control variables: the natural log of total MSA population in 2000, share of the total MSA population who is Black only or Hispanic only, share of total MSA households with annual household income less than $20,000, and share of total MSA households with annual household income at least $60,000.[6]

[6] I use year-2000 data instead of ACS 5-year 2012-2016 data due to challenges with crosswalking data from 2012-2016 boundaries to 2000 boundaries. Since I'm not concerned with a causal interpretation for these control variables, I would prefer 2012-2016 estimates for all variables to more precisely capture potential sources of variation in segregation outside of land use regulation but would need a more precise crosswalk.

MSA-level ACS 2016 5-year Gini control variable: The Gini measure of MSA income inequality comes from ACS 5-year 2016 MSA-level data.

MSA-level number of municipal jurisdictions in 2002 control variable: I obtain a list of local government jurisdictions from the 2002 Census of Governments (COG) and create a count variable equal to the number of municipalities (cities, townships, etc.), which excludes county, school district, and other non-general-purpose municipal governments. Joint city/county governments like San Francisco count only once, and separate county governments don't count at all. I attempt to follow Rothwell and Massey (2009, 2010) and Lens and Monkonnen (2016) in this definition. The 2002 COG file identifies a municipality's county but not MSA, so I use the Missouri Census Data Center's Geocorr program to match year-2000 counties to year-2000 MSAs and PMSAs.

3.3.4 Instrumental Variables

MSA-level 1982 local public protective inspection and regulation expenditures as a share of total local public revenues: I use 1982 Census State and Local Government Finance Data at the individual jurisdiction level and crosswalk these year-1982 jurisdictions into year-2000 MSAs. Dividing MSA-level total protective expenditures by MSA-level total revenues produces a protective expenditure share variable.

MSA-level number of streams: Following Dawkins (2005), I join the 1991 USGS 48-state hydrography layer from 1:2,000,000-scale DLG data to NHGIS 2000 MSA boundary data in ArcGIS Pro. For each MSA, I count the number of stream segments at least 1km in length.

4 Results

4.1 Information Theory Index of Income Segregation

4.1.1 Wharton Residential Land Use Regulation Index (WRLURI)

The main regression table shows economically small negative effects of WRLURI on the Information Theory Index measure of household income segregation. They are statistically signficant (<1% level) across all spatial scales for estimates with IVs (and for all OLS except just 5% level for 500m and 8000m radii). Aspatial estimates (columns 1 and 2) were only marginally significant at 10% level. IV estimates roughly doubled the OLS coefficient estimate magnitudes. A standard deviation increase in WRLURI is associated a 0.31 of standard deviation decrease in the aspatial Theil index of household income segregation. For spatial versions of the index, a decrease of 0.58 of a standard deviation for the 500m radius, 0.56 for 1000m, 0.53 for 2000m, 0.50 for 4000m, and 0.43 for 8000m.

This implies an increase in restrictiveness of land use regulation in St. Louis MSA from 25th percentile

to 75th percentile (right above Sacramento and right below Riverside) would move segregation from the 78th percentile to the 49th percentile for the 500m radius (just below West Palm Beach but slightly above Tampa), 63rd to 27th percentile for 1000m (below Newark but still above Ft. Lauderdale), 74th to 47th percentile for 2000m (just below Washington and just above Norfolk), 86th to 70th percentile for 4000m (just below Cincinnati but still above Washington), and 90th to 84th percentile for 8000m (below Newark, just above New York). This translates to a 15% (500m radius), 19% (1000m), 22% (2000m), 23% (4000m), and 22% (8000m) cut in segregation in St. Louis.

Reducing regulation in Riverside MSA from 75th percentile to 25th percentile would move segregration from the 57th to the 85th percentile for the 500m radius (above Greensboro and just below Louisville), 55th to 78th percentile for 1000m (above Jacksonville but below Fort Worth), 58th to 80th percentile for 2000m (just above Indianapolis but still below Columbus), 72nd to 87th percentile for 4000m (above Jacksonville but just below Memphis and below Dallas and Forth Worth), and 83rd to 90th percentile for 8000m (above Memphis but below St. Louis). This translates to a 17% (500m radius), 20% (1000m), 26% (2000m), 28% (4000m), and 28% (8000m) rise in segregation in Riverside.

For the Theil indices of segregation by ratio of a person's income to the poverty threshold, coefficient estimates were similar but were about 20% smaller and rarely statistically significant (5% level for 4000m IV, 10% level for 500m OLS, 1000m IV, and 2000m IV), perhaps because it doesn't capture variation in income for anyone earning more than 200% of the poverty threshold.

4.1.2 Local Political Pressure Index (LPPI)

LPPI had a marginally statistically significant (10% level) negative association with all the spatial Theil indices of household income segregation but only in IV estimates, which suffer from weak instruments such that true standard errors may be larger. All other specifications showed no evidence of an effect. LPPI is especially vulnerable to endogeneity because it captures current public attitudes toward land use regulation issues and is much more likely determined by recent conditions.

4.1.3 State Political Involvement Index (SPII)

SPII had a statistically significant (1% level) negative association with all aspatial and spatial levels of Theil segregation by household income except only marginally significant at 10% level for aspatial IV regression. The IV is strong but at 500m scale is close to rejecting null of excludability with 0.06 Sargan p-value. The aspatial IV estimate was just 12% larger in magnitude than the OLS estimate, but spatial IV estimates were 62-78% larger in magnitude. With IV, a standard deviation increase in SPII is associated with a decrease of 0.25 of a standard deviation in aspatial Theil, 0.45 for spatial Theil using 500m radius,

0.48 for 1000m and 2000m, 0.46 for 4000m, and 0.40 for 8000m. SPII had a statistically significant (1% level) negative association with all Theil index segregation by poverty ratio in OLS regressions but had not significant associations in IV regressions except a marginally significant (10% level) negative effect for 500m spatial scale. Magnitudes were generally similar across spatial scales, with a standard deviation increase in SPII associated with about a 1/3 of a standard deviation decrease in Theil segregation by poverty ratio in most of these specifications.

4.1.4 State Court Involvement Index (SCII)

SCII had no statistically significant association with the Theil index of income segregation.

4.1.5 Local Zoning Approval Index (LZAI)

LZAI had a statistically significant positive association with the aspatial Theil indices of income segregation in OLS (1% level for household income, 5% level for poverty ratio). LZAI also had a significant (5% level) positive association with Theil household income segregation at 500m spatial scale in OLS (only 10% level with IVs). Household income segregation was marginally significant for both OLS and IV at 1000m spatial scale and for only IV at 4000m scale and only OLS at 8000m scale. LZAI coefficient estimates were always positive but otherwise insignificant, and IVs are weak at explaining variation in LZAI. A standard deviation increase in LZAI is associated with an increase of 0.14 of a standard deviation in aspatial Theil index of segregation by household income, 0.13 of an s.d. in aspatial segregation by poverty ratio, and 0.12 of an s.d. in 500m spatial segregation by household income.

4.1.6 Local Project Approval Index (LPAI)

LPAI had a statistically significant (5% level) association with aspatial and 500m-scale spatial Theil indices of income segregation by poverty ratio in OLS and with 500m-scale spatial Theil by household income using IV. LPAI had marginally significant (10% level) association with 1000m-scale by poverty ratio in OLS and by household income in IV, as well as with aspatial and 500m-scale by household income in OLS. IVs provide fairly weak identification with F-statistic of 4.2. Using OLS, a standard deviation increase in LPAI is associated with an increase of 0.13 (aspatial) and 0.12 (500m spatial scale) of a standard deviation in Theil by poverty ratio and 0.09 (aspatial) and 0.11 (500m spatial scale) of a standard deviation by household income. But using IV, the sign changes from positive to negative with an s.d. increase in LPAI associated with a decrease of 0.57 (aspatial) and 1.03 (500m spatial scale) of an s.d. in Theil index of segregration by household income. If IVs are valid, this suggests added project approval veto points may either passively prolong transitional periods of neighborhoods sorting from one socioeconomic group to

another or actively promote socioeconomic diversity in neighborhood residential development, particularly at smaller spatial scales, perhaps by favoring approval of mixed-income housing developments.

4.1.7 Local Assembly Index (LAI)

LAI had no statistically significant association with the Theil index of income segregation.

4.1.8 Density Restrictiveness Index (DRI)

DRI had a significant (5% level) positive association with the aspatial Theil index of household income segregation and marginally significant (10% level) positive association with Theil by poverty ratio in OLS. IV was weak and excludability was rejected in many specifications. A standard deviation increase in DRI is associated with a 0.12 of a standard deviation increase in aspatial household income segregation (0.09 of an s.d. by poverty ratio).

4.1.9 Open Space Index (OSI)

OSI had a standard deviation increase associated with a significant (5% level) increase of 0.13 of an s.d. with 1000m and 2000m spatial scale Theil by poverty ratio in OLS with a standard deviation increase associated with a 0.13 of an s.d. increase. But using IV, a standard deviation increase in OSI has a marginally significant (10% level) association with a decrease in spatial Theil by household income of 1.30 s.d (500m), 1.23 s.d. (1000m), 1.16 s.d. (2000m), 1.09 s.d. (4000m), and a 0.94 s.d. (8000m). The IV was fairly weak with an F-statistic of 2.9.

4.1.10 Exactions Index (EI)

EI had no statistically signficant association with the Theil index of income segregation.

4.1.11 Supply Restrictions Index (SRI)

SRI had a marginally significant (10% level) negative association with all spatial Theil indices of household income segregation. A standard deviation increase in SRI associates with an decrease in spatial Theil household income segregation of 1.15 s.d. (500m and 1000m), 1.12 s.d. (2000m), 1.07 s.d. (4000m), and 0.92 s.d. (8000m). This weakly suggests permit supply caps may prolong periods of neighborhood transition from one socioeconomic group to another. But the IV is fairly weak with an F-statistic of 3.1, suggesting the true standard errors may be even larger.

4.1.12 Approval Delay Index (ADI)

ADI had a significant (1% level) negative association with Theil household income segregation in all regressions except only marginally significant for aspatial IV and only significant at 5% level for 8000m spatial scale. For Theil measures by poverty ratio, ADI was significant and negative in OlS regressions at 1% level (1000m and 2000m) and 5% level (aspatial and 4000m) but only marginally significant for 500m and 8000m. Using IV, ADI was only marginally significant for 500m spatial scale Theil index of income segregation by poverty ratio. IV coefficients were 1-4 times as large as OLS estimates. An s.d. increase in ADI associates with a decrease in Theil household income segregation of 0.37 s.d. (aspatial), 0.67 s.d. (500m), 0.63 s.d (1000m), 0.59 s.d. (2000m), 0.56 s.d. (4000m), and 0.49 s.d (8000m). The IV was pretty strong with F-statistic of 13.3. This suggests project delays may prolong neighborhood transition sorting from one socioeconomic group to another.

4.2 Index of Interaction of Low Income with High Income Households

WRLURI had a significant (1% level) positive association with index of interaction of people in low income households with people in high income households in both OLS and IV specifications. A standard deviation increase in WRLURI is associated with a 0.13 standard deviation increase in interaction of people in low income households with people in high income households, implying WRLURI reduces clustering segregation. The distance-decay interaction index showed a very small positive but significant (5% level) association only in the IV specification. A standard deviation increase in WRLURI is associated with a 0.06 standard deviation increase in distance decay interaction, implying WRLURI slightly increases spatial exposure of people in low income households to people in high income households.

4.3 Information Theory Index of Black-White Segregation

WRLURI had a significant (1% level) negative association with Theil black-white segregation in all OLS regressions, but coefficients were insignificant in all IV regressions. IVs were moderately strong with F-statistic of 11.6, but excludability was rejected for the regression at 2000m spatial scale.

5 Conclusion

The regression results find a significant negative association between WRLURI and most information theory index measures of residential segregation by income but not the distance-decay interaction index. Results show evidence of a small effect but with variation across types of land use regulation and measures of regulation. The main regression table replicates Lens and Monkonnen (2016) regressions but including

291-295 MSAs instead of just the 95 largest and with dependent variables including different spatial scales and attempted instrumental variables. Main results contrast with theoretical predictions and the results of Lens and Monkonnen (2016) albeit with modest magnitudes.

Lens and Monkonnen (2016) find no significant effect of WRLURI on income segregation, but this paper's IV estimates suggest a small but significant negative relationship. Land use regulation may slow neighborhood sorting by income and may contain development to already developed areas. MSA-wide land use regulation may push population growth to less regulated MSAs but may also make it more difficult for high income residents remaining in the MSA to isolate themselves from the remaining lower income residents.

While spatial and aspatial income segregation have similar sign coefficients, variation in MSA rankings suggests income distribution across neighborhoods changes more sharply in some MSAs than in others.

Consistent with Lens and Monkonnen (2016), coefficient signs vary by regulatory subindex. Subindexes roughly proxy different regulatory components in land use. Results suggest specific land use regulations increase and decrease residential segregation by income. IVs generally performed poorly for the subindex estimates and theoretically can't be valid for more than one subindex, which limits the implications these results have for land use policy in practice. Future work should examine specific land use policies.

Why does this paper find a significant effect for the overall WRLURI? Segregation measures accounting for nearby neighborhoods may capture a closer relationship than the aspatial measure. Additionally, the IVs may help address endogeneity by which some less segregated areas may choose to adopt less restrictive land use policies. However, IVs could be invalid if past segregation patterns drive preferences for greater regulation or other local regulatory policies drive desegregatation. Similarly stream count may drive segregation directly through increased sorting across greater geographic inequality in the quality of developable land (e.g. water amenities or flood zones).

Also, Lens and Monkonnen (2016) studied only MSAs with over 500,000 people, whereas this paper nearly tripled observations by including smaller MSAs for which similar regulations may have less effect on available housing supply. Future work should examine how labor or housing demand or other metropolitan characteristics may interact with land use regulation to affect segregation.

6 Appendix

6.1 Segregation Measure Construction

Prepare for non-distance segregation measures: After merging ACS 2016 5-year block-group level data with the 2016 block group to 2000 MSA crosswalk, I create variables for number of block groups in each

MSA, total population in each MSA, MSA population below poverty, total number of households in each MSA, total number of households with income below $25,000 in each MSA, mean block-group-level median household income in each MSA, rank of a block group's median household income within each MSA, and an MSA identifier variable. Next, I create the following MSA-level segregation variables (but currently only use versions of the information theory index and interaction index for the regressions):

Dissimilarity Index for Poverty:

$$D = \sum_p [P_p * |(LP_p/P_p) - (LP/P)|]/[2 * LP * (1 - (LP/P))]$$

where D is the MSA-level dissimilarity index for poverty, p is block group p, P is population, and LP is population below the poverty threshold.

Dissimilarity Index for Household Income:

$$DHHI = \sum_p [P_p * |(LP_p/P_p) - (LP/P)|]/[2 * LP * (1 - (LP/P))]$$

where $DHHI$ is the MSA-level dissimilarity index for household income, p is block group p, P is number of households, and LP is number of households with income less than $25,000.

Exposure (of Poor to Affluent) Index:

$$X = \sum_p [(LP_p/P_p) * (HP_p/P_p)]/\sum_p [LP_p/P_p]$$

where X is the MSA-level index for exposure of poor to affluent, p is block group p, P is population, LP is population below the poverty threshold, and HP is population with income at least 200% of poverty threshold.

Exposure (of Low Income to High Income) Index:

$$XHHI = \sum_p [(LP_p/P_p) * (HP_p/P_p)]/\sum_p [LP_p/P_p]$$

where $XHHI$ is the MSA-level index for exposure of low income households to high income households, p is block group p, P is number of households, LP is number of households with income less than $25,000, and HP is number of households with income at least $100,000.

Interaction Index between Poor and Affluent:

$$I = \sum_p [(LP_p/LP) * (HP_p/P_p)]$$

where I is the MSA-level index for interaction between poor and affluent, p is block group p, P is population, LP is population below the poverty threshold, and HP is population with income at least 200% of poverty threshold.

Interaction Index between Low and High Income:

$$IHHI = \sum_p [(LP_p/LP) * (HP_p/P_p)]$$

where $IHHI$ is the MSA-level index for interaction between low and high income households, p is block group p, P is number of households, LP is number of households with income less than \$25,000, and HP is number of households with income at least \$100,000.

Neighborhood Gini Index:

$$Gini = [(N+1)/(N-1)] - [2/(N-1) \sum_p M_p] * [\sum_p (O_p * M_p)]$$

where $Gini$ is MSA-level Gini index of inequality in median household income across neighborhoods, N is number of block groups in MSA, M_p is median household income in block group p, and O_p is rank of block group p in MSA by median household income.

Rank-Order Information Theory Index (Theil's H) for Poverty: I use the STATA "rankseg" command to create a rank-order information theory index using population counts of bins for the following percentage ranges of the poverty threshold: 0-49%, 50-99%, 100-124%, 125-149%, 150-184%, 185-199%, and 200%+. I set order at zero.

Rank-Order Information Theory Index for Income: I use the STATA "rankseg" command to create a rank-order information theory index using number of household counts of bins for the following household income ranges: \$0-9,999, \$10,000-14,999, \$15,000-19,999, \$20,000-24,999, \$25,000-29,999, \$30,000-34,999, \$35,000-39,999, \$40,000-44,999, \$45,000-49,999, \$50,000-59,999, \$60,000-74,999, \$75,000-99,999, \$100,000-124,999, \$125,000-149,999, \$150,000-199,999, and \$200,000+. I set order at zero.

Information Theory Index for Race: For the simpler non-rank information theory indexes, I create my own code for a "Theil's H" index variable H, attempting to closely follow Reardon and Townsend who supply STATA with a "seg" command. Alternatively, I could just use the "seg" command but current results reflect calculations by my own STATA construction of the following equation from Reardon (2011) and Reardon, Bischoff, Owens, and Townsend (2018) for racial categories w and x:

$$H = \sum_p \{[(P_{wp} + P_{xp})/(P_w + P_x)] * (v - v_p)/v\}$$

where $v = -[c * log_2 c + (1-c) * ln_2(1-c)]\ if\ 0 < c < 1,\ 0\ else,\ c = P_w/(P_w + P_x)$, and $p = block\ group\ p\ in\ MSA$. I calculate the following race-pair indexes: White-Black, White-Asian, White-Hispanic, White-Black/Asian/Hispanic, and White-Nonwhite. White is the White non-Hispanic population, Black is the Black non-Hispanic population, Asian is the Asian non-Hispanic population, Hispanic is the population that is Hispanic/Latino, Black/Asian/Hispanic is the population that is Black, Asian, or Hispanic. Nonwhite is the entire population except those who are White non-Hispanic.

Prepare for distance segregation measures: After creating MSA-level non-distance segregation measures, I add the MSA identifier variable to a datafile from the NBER Census Block Group Distance Database and divide the NBER block group distance datafile into separate files by MSA. The NBER datafile includes the distance in miles between every pair of 2010 block groups within 50 miles of one another. NBER calculates great-circle distances using the Haversine formula from internal points in the block group (point closest to a block group's geographic center that is within the boundaries of the block group). I add all the block-group-level population count variables to each file twice (one corresponding to block group one and the other corresponding to block group two) so that each block group pair has their individual block group population counts for each poverty and income category. Next, I create the following distance-based segregation measures (but currently don't use Hodler, Valsecchi, and Vesperoni's measure in the regressions):

Distance Decay Interaction Index of Poor with Affluent:

$$DDI = \sum_p \{(LP_p/LP) * \sum_q [(P_q^{-d_{pq}} / \sum_p P_p^{-d_{pq}}) * HP_q/P_q]\}$$

where DDI is MSA-level index of distance decay interaction of poor with affluent, P is population, LP is population below the poverty threshold, HP is population with income at least 200% of poverty threshold, and d_{pq} is distance in miles between the internal points of block groups p and q.

Distance Decay Interaction Index of Low with High Income:

$$DDIHHI = \sum_p \{(LP_p/LP) * \sum_q [(P_q^{-d_{pq}} / \sum_p P_p^{-d_{pq}}) * HP_q/P_q]\}$$

where $DDIHHI$ is MSA-level index of distance decay interaction of low with high income households, P is number of households, LP is number of households with income less than \$25,000, HP is number of households with income at least \$100,000, and d_{pq} is distance in miles between the internal points of block groups p and q.

Hodler-Valsecchi-Vesperoni Index of Spatial Income Segregation:

$$HVV = \sum_{p,q} \sum_{m,n} (P_{pm}/P_p) * (P_{qn}/P_q) * \lambda_{pq} * \gamma_{mn}$$

where HVV is MSA-level index of spatial income segregation using segregation formula from Hodler, Valsecchi, and Vesperoni (2017); λ_{pq} is the normalized distance between the internal points of block groups p and q; and γ_{mn} is the normalized dollar-distance between the midpoints of income bins m and n.

Spatial Rank-Order Information Theory Indexes for Poverty and for Income: Following Reardon et al (2006), I create separate indexes that account for the population composition of nearby block groups up to each of the following radii r: 500m, 1000m, 2000m, and 4000m. The "rankseg" command doesn't include a spatial option, so I create spatial-adjusted rank-order bins. First, I create the biweight proximity function: $w_{pq} = (1 - (d_{pq}/r)^2)^2 \; if \; d_{pq} < r, \; 0 \; else$, for every combination of block groups p and q. For each group m in each block group p, I create the spatially-weighted local population composition (LPC) function: $LPC_{pm} = \int P_{qm} * w_{pq} dq / \int P_q * w_{pq} dq$, that is I divide the sum of all these weighted nearby-block-group populations. I use the sum for all block groups p of $P_{qm} * w_{pq}$ in place of the aspatial population count bins and run "rankseg" command by MSA with order zero to create MSA-level spatial rank-order information theory indexes.

Spatial Information Theory Indexes for Race: Similarly to Spatial Rank-Order Information Theory Indexes above, I follow same procedure to calculate spatial versions of the IT measures of racial segregation above using the spatial-weighted local population composition (LPC) function's $P_{qm} * w_{pq}$ in place of the aspatial racial population counts in the IT equation.

6.2 Control Variable Construction

MSA-Level US Census 2000 Control Variables: From the decennial US Census 2000 SF1 and SF3 data files at the PMSA/MSA level, I obtain the following metropolitan-level control variables: the natural log of total MSA population in 2000, share of the total MSA population who is Black only or Hispanic only, share of total MSA households with annual household income less than $20,000, and share of total MSA households with annual household income at least $60,000.[7]

MSA-level ACS 2016 5-year Gini control variable: The GINI measure of MSA income inequality comes from ACS 5-year 2016 MSA-level data. I use the Missouri Census Data Center's Geocorr program to crosswalk CBSA-level GINI Index of Inequality from ACS 2016 5-year to year 2000 MSAs and PMSAs

[7] I use year-2000 data instead of ACS 5-year 2012-2016 data due to challenges with crosswalking data from 2012-2016 boundaries to 2000 boundaries. Since I'm not concerned with a causal interpretation for these control variables, I would prefer 2012-2016 estimates for all variables to more precisely capture potential sources of variation in segregation outside of land use regulation but would need a more precise crosswalk.

using population-based weights.

MSA-level number of municipal jurisdictions in 2002 control variable: Jurisdictional fragmentation likely increases income segregation. Fragmenting municipal services within an MSA will increase the incentive for high income to self sort into more expensive neighborhoods by exacerbating neighborhood quality differences. It also will encourage high income municipalities to restrict housing available for lower income families to avoid free riding by low income families and to capture rent from local housing demand. I obtain a list of local government jurisdictions from the 2002 Census of Governments (COG) and create a count variable equal to the number of municipalities (cities, townships, etc.), which excludes county, school district, and other non-general-purpose municipal governments. Joint city/county governments like San Francisco count only once and separate county governments don't count at all. I attempt to follow Rothwell and Massey (2009, 2010) and Lens and Monkonnen (2016) in this definition. The 2002 COG file identifies a municipality's county but not MSA, so I use the Missouri Census Data Center's Geocorr program to match year-2000 counties to year-2000 MSAs and PMSAs.

6.3 IV Construction

MSA-level 1982 local public protective inspection and regulation expenditures as a share of total local public revenues: Saiz (2010) uses this variable to instrument for WRLURI in its regressions on housing supply elasticity.[8] I adopt it for my segregation regressions under the assumption that protective inspection and regulation spending only affects segregation through land use regulation. To create the protective inspection and regulation variable, I use 1982 Census State and Local Government Finance Data at the individual jurisdiction level and crosswalk these year-1982 jurisdictions into year-2000 MSAs.[9]

[8]Saiz (2010) explores the determinants of housing supply elasticity and estimates housing supply elasticities of different MSAs. His conceptual model of city housing supply predicts that price sensitivity to demand shocks and productivity and amenities decrease with land availability and that population size is independent of land availability. He runs a 3SLS with IVs for log WRLURI (1982 log protective inspection and regulation spending share of public revenues, 1971 nontraditional Christian share, and 1940-1970 city decline dummy) and for housing quantity (hours of sun, immigration shocks, and Bartik employment shift-share and interaction with share of undevelopable geography). Share of undevelopable land is negatively associated with housing supply elasticity and positively associated with housing price levels and growth, 1970-2000, controlling for regional effects and coastal markets. Share of undevelopable land and WRLURI are closely and positively associated and can describe housing supply elasticity. Log of local public expenditures on protective inspection and regulation as a share of total public revenues is strongly positively associated with log WRLURI. Non-traditional Christian share of population is strongly negatively associated with WRLURI (presumably via evangelical emphasis on individualism and limited government). Geographic constraints in growing cities and higher housing prices contributed to 2005 land use regulation. I adopt the 1982 log protective inspection and regulation spending share of public revenues to instrument for WRLURI in my segregaation regressions on the assumption that this variable only affects segregation through its association with land use regulation.

[9]1982 Census State and Local Government Finance Data at the individual jurisdiction level comes from https://www.census.gov/programs-surveys/gov-finances/data/historical-data.html. I crosswalk changes in geographic GOVS ID over time and convert from 1982 GOVS ID to 2002 FIPS county code using Census of Governments files: (IDxWalk.txt and GOVS_to_FIPS_Codes_State_&_County_2007.xls). I adjust 2002 FIPS county codes for certain governments using David Dorn's list of changes since 1982 to reflect county boundaries in 2000 (https://www.ddorn.net/data/FIPS_County_Code_Changes.pdf). Otherwise, I assume county boundaries are unchanged between 1982 and 2002. Missouri Census Data Center provides a 2000 county to 2000 PMSA and 2000 MSACMSA crosswalk

After merging the 1982 state and local government finance data with the crosswalk, I collapse the dataset by 2000 MSA, creating variables for total protective inspection and regulation expenditures and total revenues by all local government jurisdictions in 1982 within the 2000 MSA counties, using county to MSA land area weights. Dividing msa-level total protective expenditures by msa-level total revenues produces a protective expenditure share variable. I use the log of protective expenditure share as an instrumental variable in my primary specifications.

MSA-level 1910 population density: NHGIS provides county population and area data from the 1910 Census. To crosswalk to 2000 MSA, I convert geographic identifiers to merge the data with the Eckert, Gvirtz, and Peters (2018) 1910-1990 county crosswalk. I assume 1990 counties comprising at least 50% of 2000 counties are identical to those 2000 counties. Missouri Census Data Center matches 2000 counties to 2000 PMSAs and MSAs, using area weights. After combining the data sets, I divide county 1910 population by county 1910 area in square miles to create a county 1910 population density variable and then collapse the dataset by MSA (PMSA and MSA) to create a 2000 MSA-level variable of the area-weighted mean 1910 population density among counties composing the MSA. 1910 population density is one of two instrumental variables used in an alternative specification paralleling Rothwell and Massey (2010).

2016 Block Group to 2000 MSA crosswalk: Missouri Census Data Center matches every 2000 Census block to its corresponding 2000 Census PMSA or MSA. NHGIS crosswalks 2000 Census blocks to 2010 Census blocks.

ACS 2016 5-Year Block-Group-level data file: I create a single data file with ACS 2016 5-year block-group level data from all 50 states and DC, including geographic, population, housing, income, and race variables. I merge ACS 2016 5-year block-group level data with the 2016 block group to 2000 MSA crosswalk. Block-group-level median household income is top coded at $250,000 and usually bottom coded at $2,500. I assume all block groups with median household income $250,000 or greater equal $250,000 and with $2,500 or less equal $2,500. For block groups with alternative bottom coding, I similarly assume the median household income equals the bottom code.

Statehood year: The year of statehood instrumental variable comes from the US mint website. After merging ACS 2016 5-year block-group level data with the 2016 block group to 2000 MSA crosswalk, I create a variable equal to the year of statehood for the state containing the block group. Collapsing the variable by MSA creates a block-group-number-weighted average statehood year variable at the MSA level. Statehood year is one of two instrumental variables used in an alternative specification paralleling Rothwell and Massey (2010). Statehood year data is obtained from the US Mint.

with land area weights (http://mcdc.missouri.edu/applications/geocorr.html). This paper defines 2000 MSAs as all 2000 PMSAs and MSAs (CMSAs are combinations of PMSAs and are thus ignored).

Dillon's Rule: I also create a dummy variable equal to "1" if the state containing the block group at least partly upholds Dillon's Rule (a judicial principle from 1873 that local governments only have authority over what states expressly authorize), and "0" otherwise.[10] Richardson, Gough, and Puentes (2003) identify states applying Dillon's Rule. I denote all states as Dillon's Rule states except for: Alaska, Iowa, Massachusetts, Montana, New Jersey, New Mexico, Ohio, Oregon, South Carolina, and Utah. Florida's Home Rule Powers Act of 1973 eliminated Dillon's rule except in taxation, but I count it as a Dillon's rule state. Dillon's rule applies only in Alabama counties; all California localities except charter cities; Colorado statutory cities and towns and all counties; Illinois non-home-rule municipalities; Indiana townships; Kansas school districts; townships; and special districts (not cities or counties); Louisiana pre-1974 charter municipalities; and Tennessee non-home-rule municipalities. Collapsing the variable by MSA creates a continuous MSA-level variable between 0 and 1 reflecting the share of block groups in the MSA in a Dillon's rule state. Dillon's rule offers an alternative IV paralleling Dawkins (2005).

MSA-level number of streams instrumental variable: Following Dawkins (2005), I join the 1991 USGS 48-state hydrography layer from 1:2,000,000-scale DLG data to NHGIS 2000 MSA boundary data in ArcGIS Pro. I export the attribute table identifying the list of stream segments in each MSA to a CSV file and import the data to STATA. For each MSA, I count the number of stream segments at least 1km in length. I also calculate the total length of all stream segments within each MSA but currently only use the stream count variable.

6.4 Preparing Regressions

I combine all these variables into a single data file and regress various measures of segregation on the WRLURI and its components, using ln(2000 population), 2016 GINI, share of households with income <$20,000 in 2000, share of households with income $60,000+ in 2000, share of population in 2000 who is

[10] Dawkins (2005) examines how local government fragmentation affects residential segregation by race across neighborhoods and jurisdictions. He uses 1980-2000 US Census of Population and Housing and USGS data on number of rivers. He expects higher income households will offer higher bids for housing in jurisdictions with more local public services if marginal valuation of public services rises faster with income than does demand for housing. Tiebout choice should thus increase income sorting across jurisdictions, and, given black-white income/wealth gap, increase racial sorting across jurisdictions. Tiebout choice may also increase racial segregation due to race-correlated public services preference heterogeneity and preferences for living near people of same race or near people of advantaged race. He runs a 2SLS region fixed effects regression for an MSA-level Gini segregation measure on a Herfindahl index of population shares in each jurisdiction to measure Tiebout choice instrumented by (1) whether state adopted Dillon's rule restricting local control over incorporation, fiscal policy, and legislation; and (2) number of rivers and streams. A 10% increase in Tiebout choice increases segregation across neighborhoods by no more than 1% and across jurisdictions by 4-7%. I adopt the number of rivers measure to instrument for land use regulation. If its effect on income segregation just runs through Tiebout choice as assumed in this paper, it should be valid for instrumenting land use regulation, controlling for number of jurisdictions. But number of rivers may aid segregation even in absence of land use regulation or make historic restrictions on residential location more difficult to overcome due to lower social interaction, perhaps resolvable via controls. Dillon's rule, originating in 1873 Iowa, rapidly adopted by other state courts and bolstered by the US Supreme Court in 1907 and 1923, can directly affect local discretion to set land use regulations but may be associated with other state and local policies influencing segregation. Richardson, Gough, and Puentes (2003) argues "judicial attitudes and constitutional or legislative priorities" largely randomly determine whether a state holds to Dillon's Rule.

Black only or Hispanic only, and number of municipal jurisdictions in 2002. To instrument for the land use regulation variables, I use number of streams from the 1991 USGS hydgrography layer and 1982 local public protective inspection and regulation expenditures as a share of total local public revenues.

References

[1] Albouy, David and Gabriel Erhlich (2018) "Housing productivity and the social cost of land-use restrictions" Journal of Urban Economics 107 101-120.

[2] Amior, Michael and Alan Manning (June 2015). "The Persistence of Local Joblessness." Working Paper. Centre for Economic Performance Discussion Paper No 1357.

[3] Andersson, Fredrik and Tom Mayock (2014). "How does home equity affect mobility?" Journal of Urban Economics 84 (2014) 23-39.

[4] Bartik, Timothy J. (1991). Who Benefits from State and Local Economic Development Policies? Kalamazoo, MI: W.E. Upjohn Institute for Employment Research.

[5] Berry, Christopher (2001) "Land Use Regulation and Residential Segregation: Does Zoning Matter?" American Law and Economics Review.

[6] Bischoff, Kendra and Sean F. Reardon (2013) "Residential Segregation by Income, 1970-2009" Russell Sage Foundation: John R. Logan, The Lost Decade? Social Change in the U.S. after 2000.

[7] Bricker, Jesse and Brian Bucks (2016). "Negative home equity, economic insecurity, and household mobility over the Great Recession." Journal of Urban Economics 91 (2016) 1-12.

[8] Bloze, Gintautas and Morten Skak (2016). "Housing equity, residential mobility and commuting." Journal of Urban Economics 96 (2016) 156-65.

[9] Brown, Jennifer and David A. Matsa (Dec 2016). "Locked in by Leverage: Job Search During the Housing Crisis." NBER Working Paper 22929.

[10] Chetty, Hendren, Kline, and Saez (2014)

[11] Chetty, Steppner, Abraham, Lin, Scuderi, Turner, Bergeron, and Cutler (2016)

[12] Chetty, Friedman, Hendren, Jones, and Porter (2018)

[13] Chetty, Raj and Nathaniel Hendren (2018) "The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates." Forthcoming, Quarterly Journal of Economics.

[14] Coulson, N. Edward, and Paul L.E. Grieco (2013). "Mobility and mortgages: Evidence from the PSID." Regional Science and Urban Economics 43 (2013) 1-7.

[15] Dawkins, Casey J. (2005). "Tiebout choice and residential segregation by race in US metropolitan areas, 1980-2000" Regional Science and Urban Economics 35 734-755.

[16] Ewing, Hamidi, Grace, and Wei 2016.

[17] Ferreira, Fernando, Joseph Gyourko, and Joseph Tracy (2010). "Housing busts and household mobility." Journal of Urban Economics 68 (2010) 34-45.

[18] Galster, George and Patrick Sharkey (2017) "Spatial Foundations of Inequality: A Conceptual Model and Empirical Overview" RSF: The Russell Sage Foundation Journal of the Social Sciences 3(2): 1-33.

[19] Ganong, Peter and Daniel Shoag (2017). "Why has regional income convergence in the U.S. declined?" Journal of Urban Economics 102 (2017) 76-90.

[20] Glaeser, Edward and Joseph Gyourko (2005). "Urban Decline and Durable Housing." Journal of Political Economy 2005 113:2, 345-75.

[21] Glaeser, Edward and Joseph Gyourko (2018). "The Economic Implications of Housing Supply." Journal of Economic Perspectives 32(1): Winter 2018, 3-30.

[22] Graif and Matthews 2017.

[23] Hilber, Christian A.L. (2010). "New housing supply and the dilution of social capital." Journal of Urban Economics 67 (2010) 419-37.

[24] Hipp and Boessen 2013.

[25] Hodler, Roland, Michele Valsecchi, and Alberto Vesperoni (Nov 2017) "Ethnic Geography: Measurement and Evidence" CESifo Working Papers

[26] Iceland, John, Daniel H. Weinberg, and Erika Steinmetz (Aug 2002) "Racial and Ethnic Residential Segregation in the United States: 1980-2000" Census 2000 Special Reports https://www.census.gov/hhes/www/housing/housing_patterns/front_toc.html.

[27] Intrator, Jake, Jonathan Tannen, and Douglas S. Massey (Aug 2016) "Segregation by race and income in the United States 1970-2010" Social Science Research 60, 45-60.

[28] Jargowsky, Paul A. (2018) "The Persistence of Segregation in the 21st Century," 36 Law & Inequality.

[29] Jargowsky, Paul A. and Christopher Wheeler (November 2017) "Economic Segregation in US Metropolitan Areas, 1970-2010." Paper Prepared for the 21st Century Cities Initiative.

[30] Kline, Patrick, and Enrico Moretti (Nov 2013) "People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development." NBER WP 19659.

[31] Krivo, Byron, Calder, Peterson, Browning, Kwan, and Lee 2015.

[32] Laamanen, Jani-Petri (2017). "Home-ownership and the Labour Market: Evidence from Rental Housing Market Deregulation." Labour Economics 48 (2017) 157-67.

[33] Lee, Barrett A., Chad R. Farrell, Sean F. Reardon, and Stephen A. Matthews (2019) "From Census Tracts to Local Environments: An Egocentric Approach to Neighborhood Racial Change." Spatial Demography 7:1-26.

[34] Lee, Barrett A., Sean F. Reardon, Glenn Firebaugh, Chad R. Farrell, Stephen A. Matthews, and David O'Sullivan (2008) "Beyond the Census Tract: Patterns and Determinants of Racial Segregation at Multiple Geographic Scales" American Sociological Review, Vol. 73 (October: 766-791).

[35] Lens, Michael C. (2017) "Measuring the geography of opportunity" Progress in Human Geography 2017, Vol. 41 (1) 3-25.

[36] Lens, Michael and Paavo Monkkonen (2016) "Do Strict Land Use Regulations Make Metropolitan Areas More Segregated by Income?" Journal of American Planning Association.

[37] Lester, T. William and Daniel A. Hartley (2014). "The long term employment impacts of gentrification in the 1990s." Regional Science and Urban Economics 45 (2014) 80-9.

[38] Massey, Douglas S. and Nancy A. Denton (Dec 1988) "The Dimensions of Residential Segregation" Social Forces, Vol. 67, No. 2, 281-315.

[39] Massey, Douglas S., Jonathan Rothwell, and Thurston Domina (2009) "The Changing Bases of Segregation in the United States" The Annals of the American Academy.

[40] Massey, Douglas S. and Rugh (2014) "Segregation in Post-Civil Rights America: Stalled Integration of End of the Segregated Century?" Du Bois Review 11(2): 205-232.

[41] Meltzer, Rachel (2013) "Do Homeowners Associations Affect Citywide Segregation? Evidence from Florida Municipalities" Housing Policy Debate, 23:4, 688-713.

[42] Meltzer, Rachel and Pooya Ghorbani (2017). "Does gentrification increase employment opportunities in low-income neighborhoods?" Regional Science and Urban Economics 66 (2017) 52-73.

[43] Owens, Ann, Sean F. Reardon, and Christopher Jencks (August 2016) "Income Segregation Between Schools and School Districts." American Educational Research Journal, Vol 53, No. 4, pp. 1159-1197.

[44] Reardon, Sean F. (2011) "Measures of Income Segregation." Stanford CEPA.

[45] Reardon, Sean F., Glenn Firebaugh, David O'Sullivan, and Stephen Matthews (Aug 2006) "A New Approach to Measuring Socio-Spatial Economic Segregation" Paper Prepared for the 29th General Conference of The International Association for Research in Income and Wealth.

[46] Reardon, Sean F., Lindsay Fox, and Joseph Townsend (July 2015) "Neighborhood Income Composition by Household Race and Income, 1990-2009" Annals, AAPSS, 660.

[47] Reardon, Sean F., Kendra Bischoff, Ann Owens, and Joseph B. Townsend (2018) "Has Income Segregation Really Increased? Bias and Bias Correction in Sample-Based Segregation Estimates" Demography 55:2129-2160.

[48] Reardon, Sean F. and Kendra Bischoff (2011) "Income Inequality and Income Segregation" AJS Volume 116 Number 4 (January 2011): 1092-1153.

[49] Reardon, Sean F., Stephen A. Matthews, David O'Sullivan, Barrett A. Lee, Glenn Firebaugh, Chad R. Farrell, and Kendra Bischoff (Aug 2008) "The Geographic Scale of Metropolitan Racial Segregation" Demography, Volume 45-Number 3, 489-514.

[50] Rothwell, Jonathan T. (Aug 2012) "The Effects of Racial Segregation on Trust and Volunteering in US Cities" Urban Studies 49(10) 2109-2136.

[51] Rothwell, Jonathan T. and Douglas S. Massey (2009).

[52] Rothwell, Jonathan T. and Douglas S. Massey (2014) "Geographic Effects on Intergenerational Income Mobility" Economic Geography 91(1):83-106.

[53] Rothwell, Jonathan T. and Douglas S. Massey (2010). "Density Zoning and Class Segregation in U.S. Metropolitan Areas" Social Science Quarterly 91(5): 1123-1143.

[54] Royuela, Vicente and Miguel Vargas (2010) "Residential Segregation: A Literature Review" Working Papers 7, Facultad de Economía y Empresa, Universidad Diego Portales.

[55] Saiz, Albert (Aug 2010). "The Geographic Determinants of Housing Supply" The Quarterly Journal of Economics.

[56] Sharp, Gregory and John Iceland (2013) "The residential segregation patterns of whites by socioeconomic status, 2000-2011" Social Science Research 42 1046-1060.

[57] Troesken and Walsh (2019).

[58] Velez and Richardson (2012).

[59] Weinberg, Iceland, and Steinmetz (2004) "Measurement of Segregation by the U.S. Bureau of the Census in 'Racial and Ethnic Residential Segregation in the United States: 1980-2000'".

[60] Wheeler, Christopher and Paul A. Jargowsky (June 2018) "Promoting Inclusive Communities: How Cities Can Utilize Local Housing Policy to Combat Economic Segregation".